Mustafa Burak BAŞAR

Portfolio

Documents

LinkedIn

Transcript

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GitHub

Professional Summary

- Electrical and Electronics Engineering student with hands-on experience gained through industrial internships and multidisciplinary engineering projects. Strong analytical, problem-solving, and teamwork skills with an interest in research, innovation, and engineering design. Seeking to contribute to projects in defense, aerospace, energy, and advanced technologies.

Education

- **İzmir Democracy University, İzmir, Turkey**
  Bachelor of Science in Electrical and Electronics Engineering (100% English)
  GPA: 3.02 / 4.00
  Focus Areas: Deep Learning (CNN), Data Mining, Computer Vision, Image Processing, DSP, Renewable Energy, Power Systems, SCADA, Machine Learning, Signal and Embedded Systems.

Technical Skills

Programming	Python, C, C++, Embedded C, SQL, VHDL
AI & Computer Vision	PyTorch, TensorFlow, OpenCV, YOLOv8, Scikit-Learn, MediaPipe, Deep Learning,Albumentations, NumPy, Pandas, Matplotlib
Embedded & Robotics	STM32, ESP32, Raspberry Pi, Arduino, Jetson Nano, Pixhawk, FPGA, ROS2, Gazebo,Microcontrollers, Sensor Integration
Industrial Systems	SCADA, Siemens S7-1200 PLC, Siemens TIA Portal
Engineering Tools	MATLAB, Simulink, KiCad, Proteus, STM32CubeIDE, Vivado, Git, GitHub, Linux (Ubuntu), Visual Studio Code,Jupyter Notebook,Pycharm
Communication Protocols	UART, SPI, I2C, CAN
CAD	SolidWorks, AutoCAD, SketchUp

Experience

Internships

- **Kalyon Energy**, Karapınar, Turkey – *PV Solar Power Plant Intern*
  Monitored SCADA systems and analyzed operational data from a utility-scale photovoltaic power plant. Assisted in troubleshooting communication and hardware faults in SCB, inverter, and monitoring systems. Performed thermal inspections, IV Curve measurements, and preventive maintenance activities while supporting TEİAŞ Active Power Limiting operations.
- **Okumuş Elektrik Engineering**, Konya, Turkey – *Electrical Engineering Intern*
  Assisted in the design and documentation of electrical engineering projects for photovoltaic power plants. Contributed to LV and MV electrical installations, prepared technical drawings, and participated in field inspections and electrical infrastructure planning.

Experience (continued)

Engineering Experience

- **İnciraltı Teknofest Team** – *Autonomous Navigation and Vision System Developer*
  Developed autonomous navigation and computer vision software for an unmanned ground vehicle (UGV). Implemented a YOLOv8-based traffic sign detection system using Python and OpenCV, and integrated Jetson Nano, Pixhawk, and MAVLink for real-time embedded AI applications.

Leadership & Activities

- **Project Development Club**, İzmir Democracy University – *Active Member*
  Contributed to the development, prototyping, and presentation of automation, embedded systems, and control engineering projects.
- **CIGRE Student Club**, İzmir Democracy University – *Member*
  Participated in technical seminars and workshops focused on power systems, renewable energy, smart grids, and SCADA technologies.

Engineering Projects

- **Autonomous Unmanned Ground Vehicle (UGV) – "Mihenk"**
  Role: Autonomous Navigation and Vision System Developer
  Designed and implemented an autonomous navigation and perception system by integrating Jetson Nano and Pixhawk platforms. Developed a YOLOv8-based traffic sign detection pipeline and optimized MAVLink communication for real-time embedded AI applications.
  Technologies: Python, OpenCV, YOLOv8, Jetson Nano, Pixhawk, MAVLink, Roboflow
- **UAV and Drone Systems Development**
  Role: Hardware Integration and Flight Mechanics
  Contributed to UAV subsystem integration, flight mechanics analysis, hardware validation, and sensor integration for autonomous aerial platforms.
  Technologies: Pixhawk, Embedded Systems, UAV Systems, Sensor Integration
- **Hybrid AI-Supported Hand Gesture Control System**
  Developed a real-time hand gesture recognition system using MediaPipe and a custom CNN architecture, achieving over 95% recognition accuracy with 180–200 ms latency.
  Technologies: Python, TensorFlow, OpenCV, MediaPipe, PyAutoGUI
- **Multi-Label Lung Disease Diagnosis**
  Trained and evaluated a DenseNet121-based deep learning model for multi-label chest X-ray classification using the NIH ChestX-ray14 dataset, achieving class-wise AUC scores up to 0.924.
  Technologies: Python, PyTorch, DenseNet121, Pandas, Albumentations
- **Image Enhancement and Restoration with U-Net**
  Developed a U-Net-based image restoration model for artifact removal, denoising, and contrast enhancement, achieving approximately 0.89 SSIM and 18 dB PSNR.
  Technologies: Python, PyTorch, U-Net, OpenCV
- **STM32-Based Robotic Arm with Memory Algorithm**
  Designed and implemented an STM32-based robotic arm with manual and autonomous operation using a memory-based motion recording and playback algorithm.
  Technologies: STM32, Embedded C, PWM, Servo Motors
- **STM32-Based Smart Irrigation System**
  Designed an automated irrigation system using an STM32 microcontroller to optimize water consumption based on soil moisture and environmental sensor data.
  Technologies: STM32, Embedded C, ADC, I2C, Sensors

Engineering Projects (continued)

- **IoT-Based Smart Parking Barrier System**
  Developed an ESP32-based smart parking barrier with ultrasonic and infrared sensors supporting automatic and manual control through a web interface.
  Technologies: ESP32, C++, IoT, Web Server
- **Glass-Fiber Reinforced Wind Turbine Design**
  Contributed to the design, 3D modeling, mold development, and manufacturing of a glass-fiber reinforced composite wind turbine.
  Technologies: SolidWorks, Composite Manufacturing, CAD Design
- **Istanbul Public Transportation Optimization**
  Analyzed Istanbul Metropolitan Municipality transportation datasets using K-Means clustering to identify peak demand periods and improve transportation efficiency.
  Technologies: Python, Pandas, Scikit-Learn, K-Means

Supporting Documentation
Detailed project reports, source code, technical documentation, certificates, and demonstration materials are available through my Portfolio Website and Google Drive links provided in the header.

Research Interests

- Artificial Intelligence, Autonomous Systems, Robotics, FPGA-Based Systems, Quantum Computing, Nuclear Energy Engineering, Renewable Energy Systems, Industrial Automation, Signal Processing

Certifications

Defense and Aerospace

- **Directorate General of Civil Aviation (DGCA):** UAV Pilot Certificate (IHA-1)
- **Defense Industry Academy:** Fundamentals of Rocketry
- **Turkish Engine Industries (TEI):** TEI Motor School
- **Ege University:** Aviation Systems Training
- **T3 Foundation:** Unmanned Aerial Vehicle (UAV) Training

Artificial Intelligence and Software

- **İş Bankası:** Artificial Intelligence Training
- **BTK Academy:** Machine Learning with Python; STM32 Embedded Software Development; AutoCAD 2D Design
- **Erciyes University:** Digital Signal Processing (DSP)
- **ODTÜ Bilgeİş:** Unity Game Development
- **Udemy:** MATLAB/Simulink Modeling and Simulation; PCB Design with KiCad; Siemens S7-1200 PLC Programming; VHDL and FPGA Design (Fundamentals)

Defense Electronics and Industrial Engineering

- **CIGRE IDU Student Branch:** Electronic Warfare Systems: Software and Hardware Testing; Industrial Importance of Calibration and Measurement

Energy and Engineering

- **IEEE Ege University Student Branch:** Photovoltaic (PV) Power Systems; Battery Technologies
- **T3 Academy:** Nuclear Energy Fundamentals; Quantum Computing Fundamentals
- **Republic of Türkiye Ministry of National Education:** Composite Product Manufacturing; Sheet Metal Part and Mold Design; Computer-Aided 3D Design